Worksheet # 5 Part A - Multiple Choice
Physics II

Name: Key Date: _____

Multiple Choice Questions 1 though 10

1) A potential (emf) can be induced in a coil of wire by

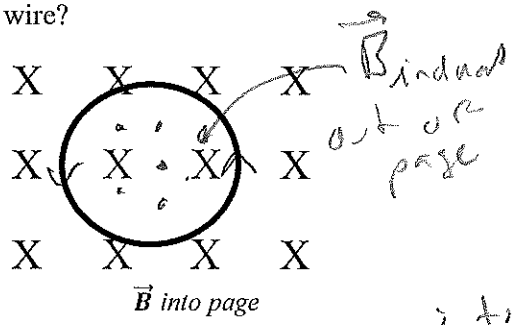
$V_{ind} = -N \frac{d\Phi}{dt}$

A) Changing the magnetic field strength.
B) Changing the angle, the coil makes to the magnetic field lines.
C) Changing the area of the coils.
(D) All of the above.
E) None of the above.

2) A wire in the shape of a circle with radius r is placed in a magnetic field $\vec{B}$ that is pointing into the page, if the magnetic field is getting stronger what direction is the induced current in the wire?

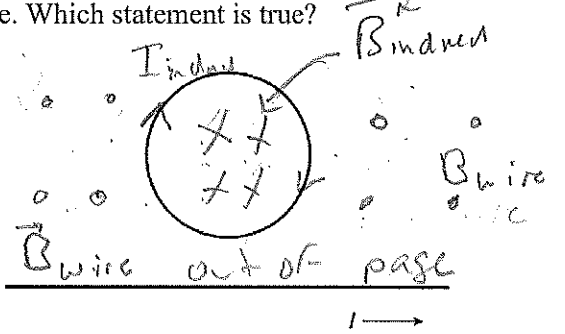
(A) counterclockwise RH Rule #2
B) clockwise
C) It could be either clockwise or counterclockwise.
D) The current in the loop must be zero.

Induced Field opposes change of external field.



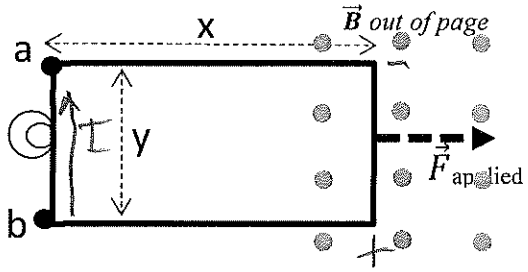
3) A circular metal ring is situated above a long straight wire, as shown in the figure. The straight wire has a current flowing to the right, and the current is increasing in time at a constant rate. Which statement is true?

(A) There is an induced current in the metal ring, flowing in a clockwise direction.
B) There is an induced current in the metal ring, flowing in a counter-clockwise direction.
C) There is no induced current in the metal ring because the current in the wire is changing at a constant rate.



4) A simple circuit consisting of a light bulb and a rectangular loop of wire that is being pulled through a magnetic field at a constant velocity as shown. What direction is the current flowing through the light bulb?

A) From a to b.
(B) From b to a.
C) Can not be determined with information given.
D) There is no induced current in this circuit.



5) A transformer changes the 10,000 V power line to 120 V. If the primary coil contains 750 turns, how many turns are on the secondary?

$\frac{N_p}{N_s} = \frac{V_p}{V_s}$ $N_s = N_p\left(\frac{V_s}{V_p}\right) = 9$ turns

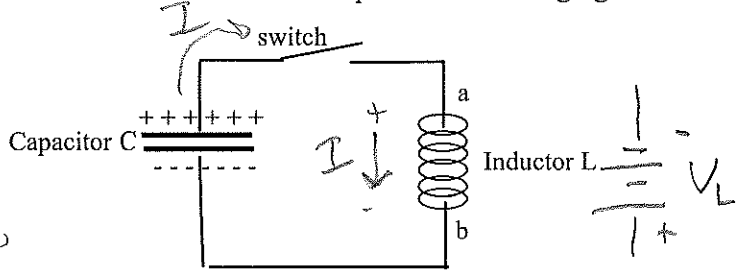
6) Which of the following statements about inductors are correct?

change in current

A) When it is connected in a circuit, an inductor always resists having current flow through it.
B) Inductors store energy by building up charge.
C) When an inductor and a resistor are connected in series with a DC battery, the current in the circuit is reduced to zero in one time constant.
D) An inductor always resists any change in the current through it.
E) When an inductor and a resistor are connected in series with a DC battery, the current in the circuit is zero after a very long time.

7) A fully charged capacitor is wire in series with a switch and an Inductor. As soon as the switch is closed, what polarity are the positions on either side of the inductor marked "a" and "b" as the capacitor is discharging?

A) a is positive and b is also positive.
B) a is negative and b is also negative.
C) a is positive and b is negative.
D) a is negative and b is positive.
E) There is no induced polarity in the inductor.

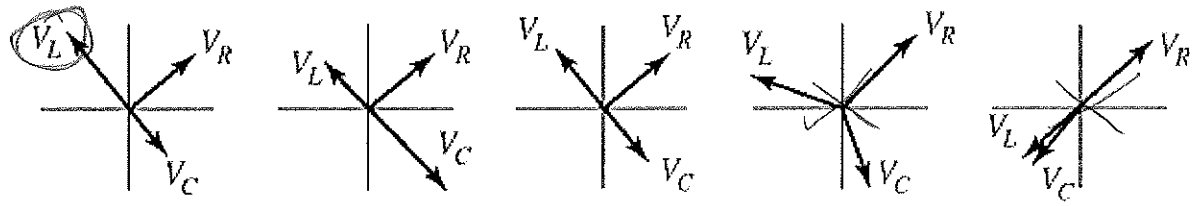


I is increasing

8) You are designing a generator to have a maximum emf of 8.0 V. If the generator coil has 200 turns and a cross-sectional area of 0.030 m^2, what should be the frequency of the generator in a uniform magnetic field of 0.03 T?

$V_{rms} = \frac{NAB2\pi f}{\sqrt{2}}$ | $f = \frac{(V_{rms})\sqrt{2}}{NAB2\pi} = 10\ Hz$

9) Which one of the phasor diagrams shown below best represents a series *LRC* circuit that is dominated by the inductance?



10) For a RCL circuit in series, determine the total impedance of the circuit if L = 10 mH, C = 100 μF, R = 10 Ω and the ac power supply is running at 120 V and 60 Hz.

$X_L = 2\pi fL = 3.8\Omega$

$X_C = \frac{1}{2\pi fC} = 26.5\Omega$

$R = 10\Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$Z = 24.8\Omega$